

Evaluation of Few-Shot Learning with Vision Language Models for Needle Bearing Defect Detection

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Abstract—Needle bearing defect detection is crucial for industrial machinery maintenance, yet traditional methods often require extensive training data and expertise. This study explores the potential of Vision Language Models (VLMs) for few-shot learning in classifying needle bearing defects using vibration signal spectrograms. We evaluate the performance of leading closed-source VLMs, including Gemini-1.5 Pro, 1.5-Flash, OpenAI GPT-4o, GPT-4o-mini, Anthropic Claude-3-5-Sonnet and Claude-3-Opus, in this task. Our results demonstrate that VLMs can effectively process and reason with spectrograms, surpassing human expert performance in needle-bearing spectrogram classification. This approach not only achieves superior accuracy but also offers a resource-efficient alternative to conventional deployment methods. The findings contribute to the rapidly evolving research area of VLM applications in spectrogram analysis and highlight the potential of few-shot learning with VLMs for industrial diagnostic tasks. Our study underscores the significance of this innovative approach in advancing predictive maintenance strategies and improving the reliability of industrial machinery.

Index Terms—Vision Language Models, Few-Shot Prompting, Spectrogram, Needle bearing, Fault Detection

I. INTRODUCTION

The manufacturing industry is experiencing a profound transformation driven by the convergence of digital technologies and artificial intelligence (AI), a revolution known as Industry 4.0. This paradigm shift is characterized by the integration of smart technologies into manufacturing processes, leading to the creation of intelligent factories and ushering in a new era of productivity [1]. At the core of this transformation is the ability to collect and process vast amounts of sensor data, enabling real-time decision-making, enhanced efficiency, and unprecedented levels of automation [2], [3]. Advancements in AI have spurred research efforts towards developing intelligent fault diagnosis systems for predictive maintenance, significantly enhancing the safety and reliability of critical infrastructure [4], [5]. These systems leverage machine learning algorithms to analyze complex sensor data, identify patterns indicative of potential failures, and predict maintenance needs before they lead to



Fig. 1. Needle Roller Bearing

costly downtime or catastrophic failures, thereby minimizing downtime and optimizing resource allocation. The potential impact of such technologies extends beyond mere operational improvements, promising to reshape the entire landscape of industrial manufacturing and maintenance strategies in the coming years.

Bearings play a crucial role in industrial and automotive applications, serving as essential components that facilitate smooth rotation, minimize wear, reduce friction, enable efficient power transmission, and support loads in various machinery and equipment. Their importance lies in their ability to enhance operational efficiency, minimize energy consumption, and extend the lifespan of mechanical systems. In modern manufacturing, bearings are integral to the performance and reliability of critical infrastructure, from automotive engines and transmissions to industrial machinery and wind turbines. When a bearing malfunction occurs, it can lead to catastrophic system breakdowns, resulting in severe operational and economic consequences. The failure of these essential elements not only disrupts the functionality of the entire mechanical system but can also pose significant safety risks and incur substantial repair costs. Given their crucial role and vulnerability, early detection and prevention of bearing failures have become paramount in maintaining the reliability and efficiency of industrial machinery. The importance of bearings in modern manufacturing and industrial processes has led to an increased focus on their reliability and

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performance. Fault detection in bearings is critical for several reasons, including preventing unexpected failures and costly downtime, enhancing overall equipment reliability and efficiency, improving safety in industrial operations, optimizing maintenance schedules, and reducing operational costs [6], [7], [8].

Among various bearing types, needle bearings have gained prominence due to their unique characteristics. Needle bearings are characterized by their long, thin cylindrical rollers, which allow for a high load-carrying capacity in a compact radial space. This makes them particularly valuable in applications where space is limited but high radial load capacity is required, such as automotive transmissions, industrial gearboxes, and aerospace systems. Needle bearings are critical components in industrial systems, where early defect detection is vital for preventing costly downtime. However, fault detection in needle bearings presents several challenges, such as limited space for sensor placement due to their compact design, high-frequency vibrations that can be difficult to isolate from background noise, complex load distributions that can mask early-stage faults and rapid progression of faults due to high stress concentrations [9]. These challenges require the development of advanced fault detection techniques specifically tailored for needle bearings.

Extensive research has focused on ball bearing defect detection using the Case Western Reserve University (CWRU) dataset. Traditional methodologies encompass advanced signal processing techniques, including Hilbert envelope spectrum analysis, and deep learning architectures such as Stacked Denoising Autoencoders [10] [11], Convolutional Neural Networks (CNNs), Hierarchical Deep Networks, Recurrent Neural Networks (RNNs), Transfer Learning, and Generative Adversarial Networks (GANs), as comprehensively reviewed in [12]. Recent advancements include the Gram Matrix-based Multi-Scale CNN (GMSCNN) [13], achieving 99.83% accuracy by mitigating irrelevant noise in vibration signals. Additionally, optimization for efficient deployment has been explored, notably with MobileNet-based time-series analysis achieving 96% accuracy [14]. However, these approaches typically necessitate intricate algorithm design, tailored to specific signal characteristics, and often involve trade-offs between accuracy and computational efficiency. This work diverges from these established paradigms by investigating the application of Vision-Language Models (VLMs) with few-shot prompting for simplified defect detection. A key innovation lies in the elimination of extensive retraining or algorithm development for transient defect identification. By leveraging spectrogram snapshots as few-shot prompts, the VLM enables operators to directly identify daily anomalies, offering a streamlined and adaptable solution compared to traditional methods.

Recent advancements in artificial intelligence, particularly in the domain of computer vision and natural language processing, have opened new avenues for fault detection. Vision Language Models (VLMs), which excel at interpreting and reasoning about visual data, show promise in this field.

The VSLLaVA pipeline [15] was created to analyse industrial vibration signals by utilising large multimodal foundation models. Through low-rank adaptation techniques, it refines CLIP and large language models by generating signal-question-answer triplets using an expert rule-assisted signal generator. This method incorporates knowledge of multimodal signal processing into the models. Using a mix of expert rules and big language model capabilities, the performance of the fine-tuned model is assessed, demonstrating increased accuracy in defect detection and signal parameter identification. Based on the LLaVA model architecture, Diagnostics-LLaVA is a VLM intended to support maintenance and repair activities in certain industrial domains. As an example of Diagnostics-LLaVA for the automobile industry, researchers developed automobile-LLaVA. In order to train a CLIP-based Vision Transformer model for integrated text and image representation using adapters, the development approach entails building an Automotive image-text pairings dataset. For supervised instruction tuning of the LLaVA model, an Automotive Visual Question Answering dataset is also generated, which consists of pictures, questions, and answers. Additionally, the researchers provide a benchmark to measure the performance of the Automotive-LLaVA model. This strategy seeks to improve VLMs' performance in specialized industrial applications, which might lead to better maintenance and repair procedures in the automobile sector [16].

Visual Spectrogram has been demonstrated for an environmental sound classification dataset [17]. In this study, we leverage the capabilities of VLMs for needle bearing fault detection. The transformation of vibration data into spectrograms creates a visual representation of frequency content over time, effectively bridging the gap between the acoustic domain of bearing vibrations and the visual domain where VLMs demonstrate exceptional proficiency. This approach enables the use of advanced visual processing capabilities inherent in VLMs for the analysis of complex auditory phenomena, potentially revolutionizing the field of bearing fault detection. Our research introduces the concept of VSC for needle bearing fault detection. This method utilizes few-shot learning techniques, allowing VLMs to classify bearing conditions with minimal training examples. We evaluate the performance of several state-of-the-art VLMs, including leading closed-source VLMs, including Gemini-1.5 Pro, 1.5-Flash, OpenAI GPT-4o, GPT-4o-mini, Anthropic Claude-3-5-Sonnet and Claude-3-Opus, on this task. [18] [19] [20].

The study includes a comprehensive analysis of various spectrogram representations using Short Time Fourier Transform and Multitaper Spectral Estimation, exploring how different visual encodings of vibration data impact classification accuracy. We also compare the performance of VLMs against the state of the art approach involving fine tuning Vision Transformers and also human expert assessments. Our findings suggest that VLMs can achieve remarkable accuracy in needle bearing fault detection, often surpassing human experts and reaches accuracy levels closer

to ViT fine tuning approach [21]. Our novel approach offers several advantages, including reduced reliance on extensive labeled datasets and the potential for more generalizable fault detection models. The implications of this research extend beyond needle bearings, suggesting new possibilities for fault detection in various industrial applications.

II. METHODOLOGY AND IMPLEMENTATION

A. Data Collection

The data collection setup closely resembled the standard setup used by the Case Western Reserve University Bearing Data Center [12]. The bearing test rig consisted of a 2 HP motor, torque converter, encoder, dynamometer, and control device. A drawn cup needle roller bearing manufactured by Schaeffler supported the motor shaft. The specific bearing model number was not disclosed. The bearing supported the motor shaft, and torque was transmitted to the shaft via the dynamometer and electronic control system. Vibration data was collected using an ICP accelerometer (M353B15) from PCB Piezotronics, mounted on the bearing housing. A mechanical load of approximately 50N was applied to the bearing, rotating at 900 RPM during testing. Lubrication was provided, and the operating temperature was maintained at a steady state. The accelerometer signals were sampled at a rate of 24 kHz and collected for 0.6 seconds. Raw vibration data was then preprocessed by applying an Anti-aliasing Finite Impulse Response (FIR) filter with a cutoff frequency of 12 kHz. This research focuses on the top three needle bearing defects commonly encountered in manufacturing: Dents, Needle Rejects (NR), and Raceway Defects (RD). To ensure a robust dataset, we conducted a two-month study within a production environment, collecting rejected bearings that defied classification by existing methods. These rejected samples were then meticulously analyzed and labeled by human subject-matter experts, providing a valuable ground truth for our validation. Accepted bearings represented a disproportionately large portion of the collected samples. The study yielded 2800 samples of no-defect bearings, 240 with dents, 237 with needle rejects (NR), and 252 with Raceway Dent defects. For this study, 200 samples per class were randomly selected to ensure uniform representation.

B. Ablation Study on Spectrogram computation

Spectrograms of time series signals can be computed with various settings and configurations. Key parameters include amplitude scaling (logarithmic or linear) and color scale (e.g., viridis or magma). A comprehensive exploration of these variations is explained in the Appendix section of [17]. This study focuses on two spectrogram estimation techniques: Short Time Fourier Transform and Multitaper spectral estimation.

- Short Time Fourier Transform STFT is a time frequency analysis technique that converts a time series signal into frequency components by applying Fourier Transform to overlapping signal segments. The time series signals are divided into multiple overlapping segments termed as windows. Each window is multiplied with a windowing

function such as the Hamming window as given in equation [1]. Fourier transform as per equation [2] to each window to obtain the amplitude spectrum. It is converted into a series of power representations and plotted into a 2D spectrogram.

$$H(n) = 0.54 + 0.46 \cdot \cos\left(\frac{2\pi}{N} \cdot n\right) \quad (1)$$

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(t)e^{-2\pi i t \xi} dt \quad (2)$$

TABLE I
STFT SPECTROGRAM PARAMETERS

Parameter	Parameter values
Sampling frequency	24000 KHz
Spectrogram bandwidth	0-12 KHz
Time samples per window	0.6 seconds
Spectrogram window length	256
Spectrogram hop length	128

- Multitaper spectrogram: MT spectral analysis constitutes a nonparametric method for spectral estimation [22]. It utilizes multiple orthogonal tapers or window functions to derive independent spectral estimates from the same signal, thereby significantly diminishing bias and variance compared to conventional methodologies. These tapers are orthogonal and are specifically engineered to enhance energy concentration within the designated frequency band, thereby resulting in enhanced spectral resolution. The final estimate is derived from the average of multiple estimates, which allows for enhanced management of the trade-off between spectral resolution bias and variance. This enhances its utility for non-stationary noise signals, such as those produced by bearing vibrational activity. The computation of multitaper estimates necessitates the consideration of three critical parameters to achieve optimal time-frequency resolution estimates and to generate a dynamic spectrogram plot.

- Window Size (N): The spectral window size is the length of the data segment or samples used for spectral estimation. This parameter determines the stationarity assumptions in the data.
- Time Half-bandwidth (TW): Time Half-bandwidth product controls the tradeoff between frequency resolution and variance reduction. Higher TW bandwidth product increases the number of Slepian tapers helps in reducing the variance of the spectral estimate leading to a smoother spectrogram but at the cost of frequency resolution.
- Number of Slepian tapers (L). Increasing the tapers leads to smoother estimates averaging the noise but at the tradeoff of frequency resolution.

$$TW = \frac{N \cdot \Delta f}{2} \quad (3)$$

$$L \ll [2 \cdot TW - 1] \quad (4)$$

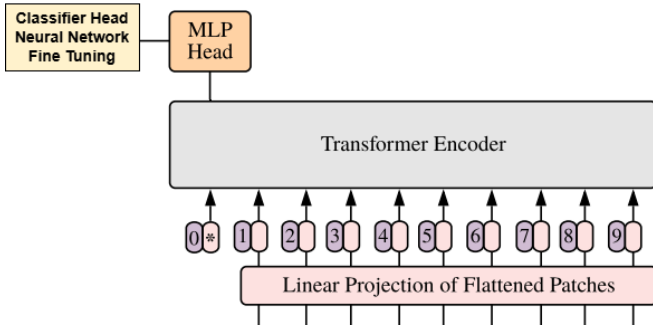


Fig. 2. ViT Model with Fine Tuning classifier head

TABLE II
MULTITAPER SPECTROGRAM PARAMETERS

Parameter	Parameter values
Sampling frequency	24000 KHz
Time BW Pdt	2.5
Num of Slepian Tapers	4
Spectrogram window length	256
Spectrogram hop length	128

C. Study Design

To evaluate the potential of few-shot learning with Vision Language Models (VLM) for the detection of needle bearing defects, an experimental methodology was designed. First, an ablation study identified the optimal spectrogram representation. The highest performing spectrogram from this experiment was then used for subsequent testings.

TABLE III
ViT FINE TUNED MODEL PARAMETERS

Parameter	Dimension	Activation functions
Embedding Layer	xx,1024	N/A
Dense Neural Network	1024,64	ReLU
Drop Out Layer	N/A	Factor = 0.5
Dense Neural network	64, NumberofClasses	softmax

A baseline study is setup with all the datasets and a ViT model as described in Figure 2 with parameters as described in Table III is used for capturing the baseline performance of a SOTA model. Early stopping with the patience parameter of 5 consecutive epochs and delta of 0.001 Adam optimizer with the learning rate of 0.001 and cosine annealing learning rate optimizer is used for fine tuning the classifier head. The performance of this SOTA model is used for comparison against the VLM with few shot prompting model. Three commercially available VLMs were tested through Application Program Interface (API) calls including two variants of one model, resulting in six models in the study. A randomly sampled data set of 100 samples per class was used for testing, iterated twice, and assembled for robust results. A temperature setting of 0 was consistently used in all models to minimize run-to-run variations. Consistent image selection across all

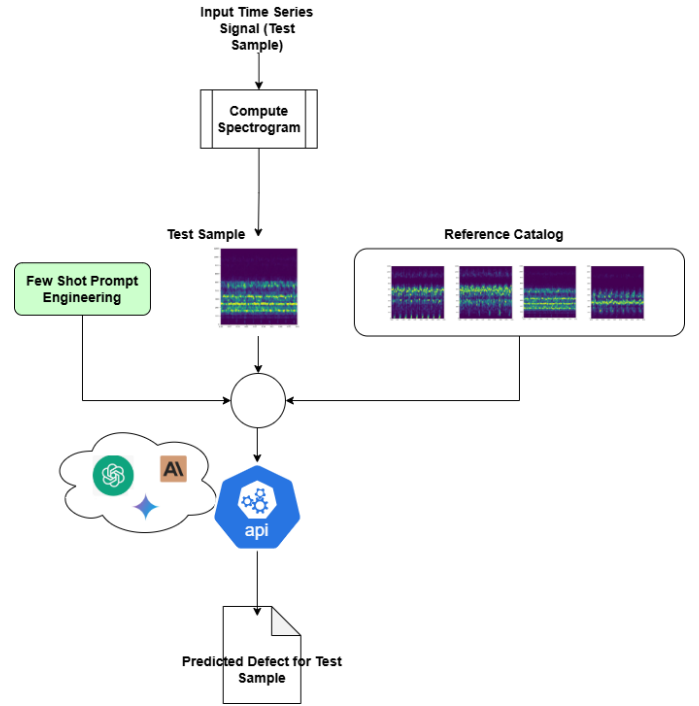


Fig. 3. Experiment Setup for Few-Shot Prompting

six models ensured an unbiased comparison. Each API call comprised a system prompt providing experimental context, a reference catalog of labeled spectrograms for a few-shot learning, the test sample spectrogram, and a classification prompt instructing the VLM. All prompts were consistent across models, with minor API-level modifications as needed. The experimental setup as illustrated in Figure [3] enabled a comprehensive and unbiased evaluation of learning with VLMs for the needle bearing defect detection. Figure [5] provides reference to the API code reference used for each of the 3 cloud sources. Choosing the few-shot Sample size K: For one-shot prompting (K=1), spectrograms from each class were converted into embeddings using the CLIP vision model, and the cluster's centroid served as the reference sample. This centroid-based approach yielded slightly improved results compared to random sample selection. For two-shot prompting (K=2), the centroid and an extreme sample from each defect class were chosen as references. Due to token length and rate limitations imposed by the study center, analysis beyond two reference samples was restricted. Prompt caching was implemented to minimize API call costs associated with repeatedly sending identical defect reference tokens. By caching these tokens, subsequent API calls only required the test image sample and the classification prompt. While this significantly improved efficiency, the 5-minute cache timeout limited the number of reference samples (K) that could be effectively managed. Consequently, this study was restricted to two reference samples per defect class. Future work will explore strategies to overcome this limitation and incorporate larger reference catalogs.

Human experts were provided with the same training samples used in the K=2 prompting scenario. They manually classified the test samples into the respective defect categories. Results from four human experts were collected, and the ensemble score served as the final evaluation.

D. Evaluation Metrics

Precision macro, Recall macro, and F1 scores are computed for each models and tabulated. Macro-averaging calculates metrics for each defect label separately and then takes the unweighted mean across all labels.

$$\text{Precision (ClassA)}_{\text{macro}} = \frac{\text{TP}(\text{ClassA})}{\text{TP}(\text{ClassA}) + \text{FP}(\text{ClassA})} \quad (5)$$

$$\text{Recall (ClassA)}_{\text{macro}} = \frac{\text{TP}(\text{ClassA})}{\text{TP}(\text{ClassA}) + \text{FN}(\text{ClassA})} \quad (6)$$

$$\text{F1}_{\text{macro}} = \frac{1}{L} \sum_{l=1}^L \frac{2 \cdot \text{Precision}(l) \cdot \text{Recall}(l)}{\text{Precision}(l) + \text{Recall}(l)} \quad (7)$$

III. RESULTS AND DISCUSSION

A. Spectrogram Study Results

Spectrogram estimation and visualization involve several parameters that can influence information perception. These include amplitude scaling (logarithmic or linear), colormap selection (e.g., jet, Viridis, Magma), frequency scaling, and the display of time and frequency axes. A detailed ablation study of spectrogram parameters for sound signals, which serves as a foundation for this work [17]. In this study, STFT and Multi-Taper estimation parameters were carefully chosen to be similar, while amplitude scaling and colormap were specifically analyzed to identify the spectrogram configuration that yielded the highest few-shot learning performance. Figure 4 attached as Appendix illustrates the different combinations of colormaps and amplitude scaling used for spectrograms generated by both STFT and Multi-Taper estimation methods. To identify the optimal configuration, 20 samples per defect class for each spectrogram variant were randomly selected and tested using one-shot prompting with the GPT-4o VLM. Table IV presents the results, demonstrating that Multi-Taper estimation with linear amplitude scaling and the Viridis colormap yielded the highest few-shot learning performance. Consequently, this configuration was selected for subsequent validation across all other VLM models.

TABLE IV
SPECTROGRAM PARAMETER STUDY

Parameter	Few Shot F1 macro
STFT Viridis Log Amplt	0.56
STFT Viridis Linear Amplt	0.74
STFT Jet Log Amplt	0.58
STFT Jet Linear Amplt	0.82
MTFT Viridis Log Amplt	0.54
MTFT Viridis Linear Amplt	0.88
MTFT Jet Log Amplt	0.52
MTFT Jet Linear Amplt	0.86

B. Human Expert Few Shot performance

Scores from human expert validation are collected and the average F1 score is estimated as a baseline to compare other VLM models.

TABLE V
FEW SHOT HUMAN PERFORMANCE STUDY

Candidate	Few Shot F1 macro
Human expert 1	0.78
Human expert 2	0.86
Human expert 3	0.82
Average Score	0.82

C. SOTA ViT model performance

SOTA ViT fine tuning model is tested for baseline comparison on VLM models.

TABLE VI
ViT MODEL PERFORMANCE

Precision macro	Recall macro	F1 score macro
0.95	0.97	0.98

D. Few Shot performance of VLM models

The initial evaluation (K=1) assessed the performance of all six VLM models in a one shot setting. In the subsequent few-shot evaluation (K=2), only the top performing models those that met or exceeded the human expert score in the initial assessment or higher parameter models were included. The basic few-shot prompting template used across all models is shown in Figure [3] of the Appendix.

TABLE VII
ONE SHOT PERFORMANCE

Model	Precision macro	Recall macro	F1 macro
GPT-4o	0.90	0.90	0.90
GPT-4o-mini	0.76	0.53	0.45
Claude-3-5-Sonnet	0.86	0.86	0.86
Claude-3-Opus	0.24	0.24	0.24
Gemini-1.5-Flash	0.54	0.34	0.25
Gemini-1.5-Pro	0.39	0.45	0.40

TABLE VIII
FEW SHOT (K=2) SHOT PERFORMANCE

Model	Precision macro	Recall macro	F1 macro
GPT-4o	0.95	0.95	0.95
Claude-3-5-Sonnet	0.94	0.94	0.94
Gemini-1.5-Pro	0.42	0.48	0.45

In comparison, GPT-4o performed well compared to other models. Anthropic Claude is much close to the performance and has a faster response time with the prompt caching feature.

TABLE IX
GPT-4o-CONFUSION MATRIX

		Predicted			
		ACPT	DENT	NR	RD
Actual	ACPT	90	10	0	1
	DENT	2	97	1	0
	NR	5	0	94	1
	RD	0	0	0	100

E. Conclusion and Further Research

This study investigated the efficacy of Vision Language Models (VLMs) in classifying needle bearing defects through the analysis of vibration signal spectrograms, a critical task for industrial machinery maintenance. Our findings unequivocally demonstrate that leading closed-source VLMs, including Gemini-1.5 Pro, 1.5-Flash, OpenAI GPT-4o, GPT-4o-mini, Anthropic Claude-3-5-Sonnet and Claude-3-Opus, can successfully process and interpret complex visual information embedded within spectrograms. Notably, these models surpass the performance of human experts in this specific classification task, approaching state-of-the-art performance (0.98 F1 score) previously achieved through traditional, resource-intensive methods [21] [24]. The fine tuning ViT model as a baseline comparison achieves F1 score of 0.98 given in Table VI. This achievement is particularly significant considering the few-shot learning approach adopted, which drastically reduces the reliance on extensive training data and specialized expertise typically required for defect detection systems. By leveraging VLMs' ability to generalize from limited examples, we have established a more efficient and accessible paradigm for industrial diagnostics.

The effectiveness of VLMs in this situation not only presents a strong substitute for conventional techniques, but it also creates promising opportunities for wider spectrogram analysis applications in a variety of industrial fields. In contrast to artificial or scholarly datasets, this study is an important first step toward methodically assessing the potential of VLMs in actual industrial settings using real-world data. The findings show that VLMs are useful instruments that can provide noticeable advantages in real-world settings rather than just being theoretical concepts. They are a popular choice for integration into the operational technology layer since they can deliver extremely precise results with minimal latency, negating the need for intensive cloud computing.

However, the present study has a few limitations. Only some of the most popular closed-source VLM models were evaluated, and it should be noted that the emerging field of open-source VLMs was not included in this study. In order to present a more thorough evaluation of the various VLM technologies, future research should address integrating these open-source alternatives. Token constraints also made it difficult to explore bigger, more representative sample dictionaries. Token limitations constrain the size and

resolution of spectrogram images that can be processed by VLMs. This could potentially affect the model's ability to detect fine-grained details in the spectrograms, which are crucial for identifying subtle bearing defects. The token limit influences the complexity and length of prompts that can be used for few-shot learning. Careful design and concise yet informative instructions are required to maximize the model's performance within these constraints. To address token limitations, preprocessing techniques such as image compression, selective cropping, or sequential processing of larger spectrograms may be necessary. These methods can help optimize the use of available tokens while preserving critical information. Token limitations may force a trade-off between capturing fine-grained spectral details and maintaining broader contextual information in the input. Balancing these aspects is crucial for accurate defect detection.

These limitations show how important it is to conduct more study on the scalability and resilience of VLMs when managing large amounts of complicated data. Additionally, although the baseline capabilities of VLMs in a few-shot situation are established in this study, it is yet unknown whether these models may be improved to perform even better on particular fault kinds or industrial conditions. The next stage of this study will concentrate on gathering a much bigger and more varied dataset in order to completely solve these shortcomings. A more comprehensive analysis of fine-tuning tactics and a more reliable assessment across a larger range of scenarios, including the previously noted token constraints, will be made possible by this larger dataset. A crucial next step is to look into how fine-tuning affects accuracy, efficiency, and adaptability.

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IV. APPENDIX

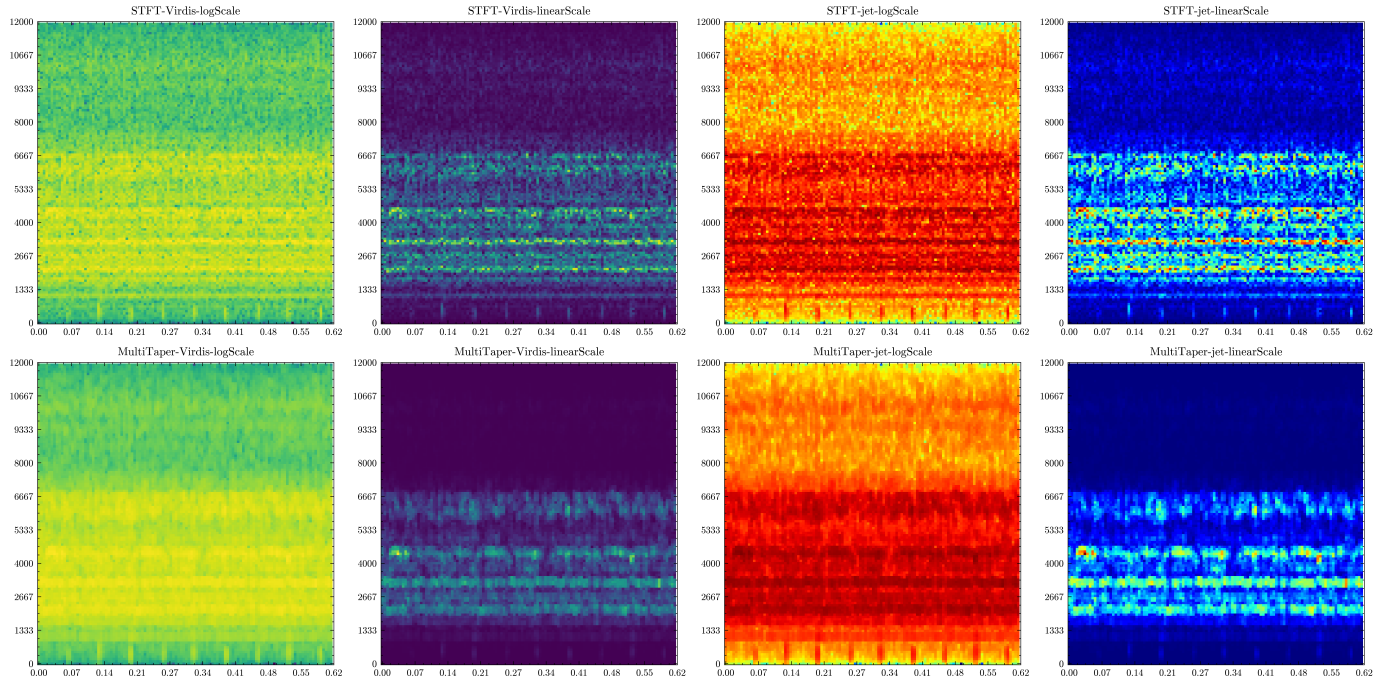


Fig. 4. Spectrogram of different configurations

```

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"content": [
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  "text": "You are an expert scientific spectrogram analyzer. Provided below are examples
  with tag ##EXAMPLE## are spectrogram corresponding to different defect classes with a
  tag ##DEFECT CLASS##. Given a plot of spectrogram images with x axis in time and y
  axis in frequency. Amplitude of the signals are color coded in standard Viridis scale.
  Extract key spectrogram information such as peak frequencies, peak amplitudes,
  frequency band characteristics, any harmonic patterns or any other relevant time
  spectrogram features. Dont hallucinate. Extract information strictly from the plot and
  associate it with the defect class"},
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  {"type": "text", "text": "##EXAMPLE## ##DEFECT CLASS: ##RACEWAY DENT"},
  {"type": "image", "source": "#base64RefImage"},

  {"type": "text",
  "text": "Now, given a new spectrogram , analyze it considering factors such as
  frequency patterns,intensity and time variations. Focus solely on the patterns
  presented in the spectrogram. Do not let any assumptions or environmental settings
  influence your decision. Your task is to determine which of the ##EXAMPLE##
  classes the new spectrogram most closely resembles . Your response must contain only
  the exact name of the class. Predicted Class:"},
  {"type": "image", "source": "base64ImageTest"}
]}

```

Fig. 5. Example of an API for Few Shot (K=1) Prompting